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Original Article

Positive End-Expiratory Pressure and Atelectasis after Bronchial Blockade One-Lung Ventilation in Infants: A Randomized Controlled Study

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Objective: This study aimed to evaluate the effect of positive end-expiratory pressure (PEEP) combined with 100% oxygen on atelectasis during emergence from anesthesia after one-lung ventilation (OLV) with a bronchial blocker in infants receiving thoracoscopic lobectomies.
Design: This was a randomized controlled study.
Setting: This study was performed in a teaching hospital.
Participants: Infants who received thoracoscopic lobectomies in our hospital from June 2024 to December 2024 were included in this study.
Interventions: Sixty-nine infants (Group A) with bronchial blocker OLV thoracoscopic lobectomies received no PEEP but maintained intraoperative oxygen levels (FiO_2 0.6-0.8) after the removal of the bronchial blocker until extubation during emergence from anaesthesia. Another 69 infants who met the inclusion criteria and received PEEP of 5 to 6 cmH_2O, combined with 100% oxygen, after removal of the bronchial blocker until extubation during emergence from anesthesia, were included in Group B.
Measurements: The inclusion criteria were as follows: (1) These infants were diagnosed with congenital cystic adenomatoid malformation and received bronchial blockade OLV thoracoscopic lobectomies; (2) American Society of Anesthesiologists (ASA) class I to II (ASA class I means a normal healthy patient; class II means a patient with mild systemic disease); and (3) age younger than 1 year. The exclusion criteria were as follows: (1) preoperative chest computed tomography or ultrasound suggestive of atelectasis; (2) intraoperative ventilation mode was changed to bilateral lung ventilation; (3) intraoperative FiO_2 greater than 0.9; (4) cardiorespiratory dysfunction; (5) large cysts causing significant mediastinal shift or compressing cardiac chambers; and (6) declined to participate.
Results: There were no significant differences in the preoperative general condition, perioperative hemodynamics, duration of operation, or length of postoperative hospital stay between the two groups ($P > 0.05$). There were no significant differences in the oxygenation index, alveolar-arterial oxygen differential pressure, or oxygen saturation (SpO_2) at T1 (the completion of surgery) ($p > 0.05$). At T2 (30 minutes after extubation), the oxygenation index and PaO_2 level were greater, the alveolar-arterial oxygen differential pressure was lower, and the lowest postoperative SpO_2 level within 30 minutes after extubation was better in Group B than in Group A ($p < 0.05$). The time from bronchial blocker removal to extubation was shorter in Group B than in Group A ($p < 0.05$). Compared with those of Group A, the proportions of patients with lung ultrasound scores of 3 (juxtapleural consolidation, large-sized consolidations and B-lines, white lung) and bedside chest radiography atelectasis scores of 3 (lobar atelectasis) in both lungs after extubation were lower in Group B ($p < 0.05$). The prevalence of hypoxemia (after extubation, $SpO_2 < 90\%$) was 46 (66.7%) in Group A and 14 (20.3%) in Group B, and the differences were statistically significant ($p < 0.05$). The duration of oxygen administration after extubation in Group B was shorter than that in Group A ($p < 0.05$). There was a statistically significant correlation between lung ultrasound scores and the chest radiographic atelectasis score in both groups ($p < 0.05$).
Conclusions: In one-lung ventilation using bronchial blockers performed during thoracoscopic lobectomies in infants, applying PEEP at 5 to 6 cmH_2O combined with 100% oxygen ventilation after the removal of the bronchial blocker until extubation was associated with improved

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oxygenation and did not appear to increase the risk of atelectasis during emergence from anesthesia. However, further studies are needed to evaluate the safety of this strategy.

Key Words: one lung ventilation; bronchial blockade; PEEP; atelectasis

CONGENITAL CYSTIC ADENOMATOID MALFORMATION (CCAM) is a rare abnormal form of lung development characterized by cystic or adenomatous overgrowth of terminal bronchioles. CCAMs are often found prenatally by ultrasound but may also present postnatally as respiratory distress or recurrent infections or may remain asymptomatic. Larger lesions can cause a significant mass effect, leading to lung compression, mediastinal shift, and even cardiac compromise. For symptomatic patients or patients at risk of malignancy, surgical resection is recommended. Currently, surgery for CCAM is recommended to be performed thoracoscopically.[1]

One-lung ventilation (OLV) is a common technique in pediatric thoracic surgery that can maintain adequate oxygenation and optimal surgical exposure.[2] Infants receiving OLV often present unique pathophysiological characteristics, which increase their susceptibility to respiratory complications during OLV. These include reduced functional residual capacity (FRC), high chest wall compliance, immature alveolar structure, and increased risk of airway collapse. These features increase the possibility of alveolar collapse, mechanical compression, and absorptive atelectasis. Atelectasis is one of the most obvious and common complications in this population, which has a significant impact on postoperative respiratory results.

Benoît et al. reported that atelectasis might also occur during emergence from anesthesia and that inhaling 100% oxygen increased the incidence of atelectasis.[3] Positive end-expiratory pressure (PEEP) is a widely recognized ventilation strategy used to prevent alveolar collapse and improve oxygenation. The application of PEEP during mechanical ventilation has been demonstrated to reduce atelectasis, enhance pulmonary compliance, and optimize gas exchange across diverse patient populations.[4] Most of the existing studies on perioperative ventilation strategies in pediatric patients have focused on intraoperative management, with limited data on the effects of PEEP and the fraction of inspired oxygen levels (FiO_2) during emergence from anesthesia. Although PEEP and 100% oxygen are widely used in extubation, the combined effects of PEEP and 100% oxygen on postoperative atelectasis in infants with CCAM following bronchial blockade OLV thoracoscopic lobectomies have still not been fully studied. Current practices are often guided by clinical preferences rather than specific evidence. Previous studies have emphasized the need for further research to determine the best extubation strategy for this special population.[5-6]

This study aimed to investigate whether a moderate PEEP level (5-6 cmH_2O) combined with 100% oxygen during emergence from anesthesia could improve oxygenation without increasing the risk of postoperative atelectasis in infants with CCAM after receiving bronchial blocking OLV during thoracoscopic lobectomies. We hypothesized that, after thoracoscopic lobectomies, infants who received PEEP of 5 to 6 cmH_2O combined with 100% oxygen, in contrast with those who received no PEEP and maintained intraoperative oxygen levels after the removal of the bronchial blocker until extubation during emergence from anesthesia, would have a lower incidence of hypoxemia and not have an increased risk of atelectasis.

The primary outcomes of this study were the postoperative atelectasis score and hypoxemia. The secondary outcome measures included the partial pressure of arterial oxygen (PaO_2), oxygenation index, alveolar-arterial oxygen differential pressure ($AaDO_2$) and oxygen saturation (SpO_2) during emergence from anesthesia; the time from bronchial blocker removal to extubation; the duration of oxygen administration after extubation; the hemodynamic parameters; and the duration of operation and postoperative hospital stay.

Materials and Methods

This study was approved by the ethics committee of our hospital. Written informed consent was obtained from the parents/guardians of the patients.

This study was conducted at a provincial tertiary-level pediatric hospital. Owing to the hospital's specialized surgical expertise and high patient volume, many infants with CCAM from various regions of the country were referred here for surgical treatment during the study period.

Infants with CCAM who received thoracoscopic lobectomies in our hospital from June 2024 to December 2024 were included in this study. Sixty-nine infants (Group A) with bronchial blocker OLV thoracoscopic lobectomies received no PEEP but maintained intraoperative oxygen levels after the removal of the bronchial blocker until extubation during emergence from anesthesia (the time from the cessation of anesthetic drugs to the point when the patient had resumed spontaneous breathing, the endotracheal tube was removed, and vital signs were stable). Another 69 infants who met the inclusion criteria and received PEEP of 5 to 6 cmH_2O combined with 100% oxygen after the removal of the bronchial blocker until extubation during emergence were included in Group B. The study design and participant flow are shown in the CONSORT diagram (Fig 1).

The inclusion criteria were as follows: (1) infants were diagnosed with CCAM and received bronchial blockade OLV thoracoscopic lobectomies; (2) American Society of Anesthesiologists (ASA) class I to II (ASA class I means a normal

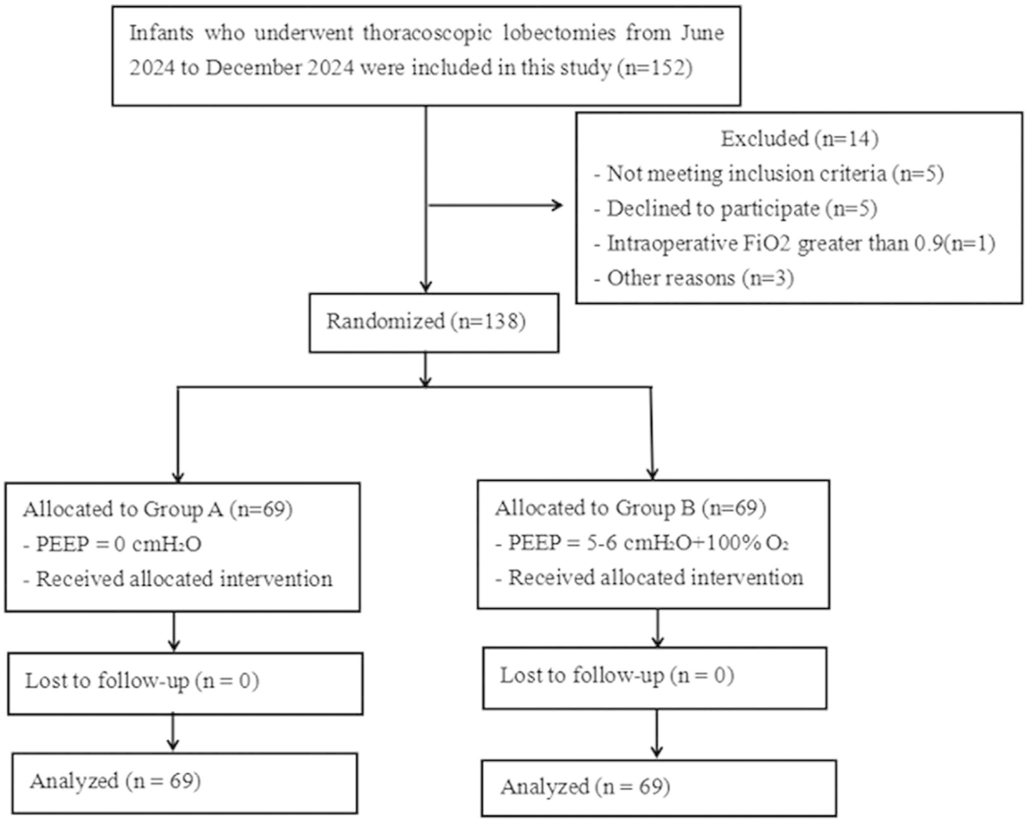

Fig. 1. CONSORT flow diagram of participant enrollment, randomization, allocation, and analysis.

healthy patient; class II means a patient with mild systemic disease); and (3) an age younger than 1 year. The exclusion criteria were as follows: (1) preoperative chest computed tomography or ultrasound suggestive of atelectasis; (2) intraoperative ventilation mode was changed to bilateral lung ventilation; (3) intraoperative FiO_2 greater than 0.9; (4) cardiorespiratory dysfunction; (5) large cysts causing significant mediastinal shift or compressing cardiac chambers; and (6) declined to participate.

Anaesthesia and Surgical Procedures

In this study, a 5 Fr pediatric bronchial blocker (with a length of 450 mm, an external diameter of 1.7 mm, and an internal diameter of 1.0 mm; Hangzhou Tanpa Medical Technology, Hangzhou, Zhejiang, China) was used for OLV. The bronchial blocker is a thin, flexible catheter with an inflatable balloon near the distal tip. The inflatable balloon has a diameter of 8 mm and is inflated with 2 mL of air. When inflated, the balloon can effectively occlude the mainstem bronchus of infants.

Upon the infant arriving in the operating room, standard ASA monitors were used (electrocardiography, blood pressure, pulse oximetry, body temperature, and end-tidal carbon dioxide partial pressure). After the induction of anesthesia, a bronchial blocker was placed through the glottic opening, rotated in the appropriate direction on the side of the intended bronchial position and advanced until resistance was reached. Then, endotracheal intubation was performed. The infants received mechanical ventilation under pressure control mode. The ventilatory parameters for both groups were as follows: FiO_2 0.6 to 0.8; inspiratory/expiratory ratio (I:E) 1:1.5; tidal volume (VT) 6 to 8 ml/kg; respiratory frequency (R) 25 to 35 times/min; fresh gas flow 2 to 3 L/min; and PEEP 3 cmH_2O. FiO_2 and respiratory frequency were adjusted according to the blood gas analysis results. A flexible bronchoscope (FB, with an external diameter of 2.0 mm; Zhejiang Youyi Medical Technology, Taizhou, Zhejiang, China) was inserted into the endotracheal tube to confirm the position of the endotracheal tube and the bronchial blocker. The endotracheal tube was adjusted to be 1 to 2 cm proximal to the carina, and the bronchial blocker was inserted into the left or right main bronchus depending on the surgical site. Arterial line placement and central line placement in the section following anesthesia induction and intubation. Arterial blood gas analyses were performed to optimize ventilation and oxygenation. The primary goals were to maintain the partial pressure of arterial carbon dioxide ($PaCO_2$) between 35 and 45 mmHg, a PaO_2 above 60 mmHg and a $SpO_2 \geq 90\%$. If SpO_2 decreased below 90% or the PaO_2 decreased below 60 mmHg, FiO_2 was gradually increased as needed. If $PaCO_2$ exceeded 45 mmHg, we increased the respiratory rate or VT accordingly. The patient was in the lateral position during the operation. All patients received lobectomies under OLV thoracoscopic surgery with bronchial blockers. OLV was performed by filling the inflatable balloon of the bronchial blocker with 1 to 2 mL of air when the skin incision was made. Upon completion of surgery, the patients were placed in the supine position. The bronchial blocker was removed, and single breath lung recruitment was performed in both groups. The pressure was 30 cmH_2O for 15 to 20 s.[7]

The infants in Group A received no PEEP but maintained intraoperative oxygen levels (FiO_2 0.6-0.8) after the removal of the bronchial blocker until extubation during emergence from anesthesia. The infants in Group B received a PEEP of 5 to 6 cmH2O, combined with 100% oxygen, after the removal of the bronchial blocker until extubation during emergence from anesthesia (a target PEEP of 5 cmH_2O was initially applied). If the peak airway pressure remained < 25 cmH_2O and oxygenation ($SpO_2 < 90\%$) required further support, PEEP was increased to 6 cmH_2O at the discretion of the attending anaesthesiologist in Group B.

When patients achieved the following criteria, we considered extubation: hemodynamic stability; $FiO_2 \leq 40\%$; peak inspiratory pressure (PIP) ≤ 18 cmH_2O; and $PaCO_2 < 50$ mmHg, $PaO_2 > 70$ mmHg, pH > 7.30, and lactic acid $<$ 2.0 mmol/L. After extubation, if $SpO_2 < 90\%$, the infant was treated with nasal cannula oxygen at 2 L/min until they maintained $SpO_2 \geq 90\%$ on room air for at least 30 minutes, which was based on common clinical practice at our institution, which provided adequate oxygenation for infants without causing discomfort. The FiO_2 value after extubation was estimated based on the oxygen flow rate using a commonly applied formula: FiO_2 (%) = 21 + (4 $\times$ oxygen flow rate in L/min). For patients receiving nasal cannula oxygen at 2 L/min, the estimated FiO_2 value was approximately 0.29. For patients who did not require supplemental oxygen, the FiO_2 value was considered to be 0.21, corresponding to room air. Patients are transferred to the intensive care unit (ICU) for further monitoring and treatment. To minimize operator differences, all anesthesia and surgical operations were performed by three cardiothoracic anesthesiologists and three pediatric thoracic surgeons in our center.

After the patients returned to the ICU, bedside chest radiography and lung ultrasound were routinely performed. The infants were examined according to the lung ultrasound localization method described by Acosta et al.[8] Juxtapleural consolidations of different sizes and the B line are the two most common lung ultrasound signs.[9] All lung ultrasound examinations were performed by an ICU physician with experience in pediatric lung ultrasound, and he was blinded to the group assignments. We recorded these two ultrasound signs and atelectasis results via chest radiography and scored the atelectasis results of both lungs.

Data were collected and subjected to statistical analysis, which included (1) patient demographic data (age, sex, and body weight); (2) the duration of the operation; and (3) mean arterial blood pressure (MAP), heart rate (HR), central venous pressure (CVP), PaO_2, oxygenation index (PaO_2/FiO_2 ratio), $AaDO_2$ ($AaDO_2 = [(FiO_2 \times (760 - 47) - (PaCO_2/0.8)] - PaO_2$, where 760 mmHg is the atmospheric pressure at sea level, 47 mmHg is the water vapor pressure at body temperature, and 0.8 is the assumed respiratory quotient) at time points T1 (the completion of surgery) and T2 (30 minutes after extubation); (4) the time from bronchial blocker removal to extubation; (5) SpO_2 at T1 and the lowest postoperative SpO_2 value within 30 minutes after extubation; (6) the prevalence of hypoxemia after extubation ($SpO_2 < 90\%$); and (7) the duration of oxygen administration; (8) lung ultrasound score after extubation of both sides of the lungs (degree of juxtapleural consolidation: 0 represents no consolidation; 1 represents minimal juxtapleural consolidation; 2 represents small-sized consolidation; 3 represents large-sized consolidation. B-lines: 0 represents fewer than three isolated B-lines; 1 represents multiple well-defined B-lines; 2 represents multiple coalescent B-lines; 3 represents white lung)[10] (Fig 2); (9) bedside chest radiography atelectasis score after extubation of both sides of the lungs (0 represents clear lung fields; 1 represents plate-like atelectasis or slight infiltration; 2 represents partial atelectasis; 3 represents lobar atelectasis; 4 represents bilateral lobar atelectasis)[11] (Fig 3); (10) postoperative hospital stay; and (11) correlation between lung ultrasound scores and chest radiographic atelectasis scores in both groups.

Statistical Analysis

The sample size was determined using PASS 11 software (NCSS, LLC, Kaysville, UT). The selection score was our primary comparing objective, so we referred to it to calculate the sample size. In the pilot study, each group included 20 infants. Among them, 8 infants (40%) in Group A and 3 infants (15%) in Group B had an atelectasis score of 2. The ratio of the study group to the control group was set at 1:1, with a significance level (alpha) of 0.05 and a statistical power of 0.90. The calculated sample size was 62 infants for the study group and 62 infants for the control group. Considering a 10% dropout rate, each group ultimately included 69 infants.

All the data were entered into Microsoft Excel and analyzed using IBM SPSS statistical software, version 20.0. Independent continuous variables were analyzed by t-tests when the data conformed to a normal distribution after testing (Shapiro-Wilk test) and are expressed as the mean $\pm$ standard deviation ($x \pm S$). The χ^2 test or Fisher exact probability method should be used for categorical data. Counts and percentages were used to describe the enumeration data. The Mann-Whitney U test was applied to non-normally distributed data. A p value less than 0.05 is considered to indicate statistical significance.

Results

Patient Demographic Data

A total of 138 infants were included in the study. There were no significant differences in the preoperative general condition (age, sex, and body weight) ($p > 0.05$) (Table 1).

Primary Outcomes

Compared with those of Group A, the proportions of patients with lung ultrasound scores of 3 (juxtapleural consolidation: large consolidation; B-lines: white lung) and bedside chest radiography atelectasis scores of 3 (lobar atelectasis) in

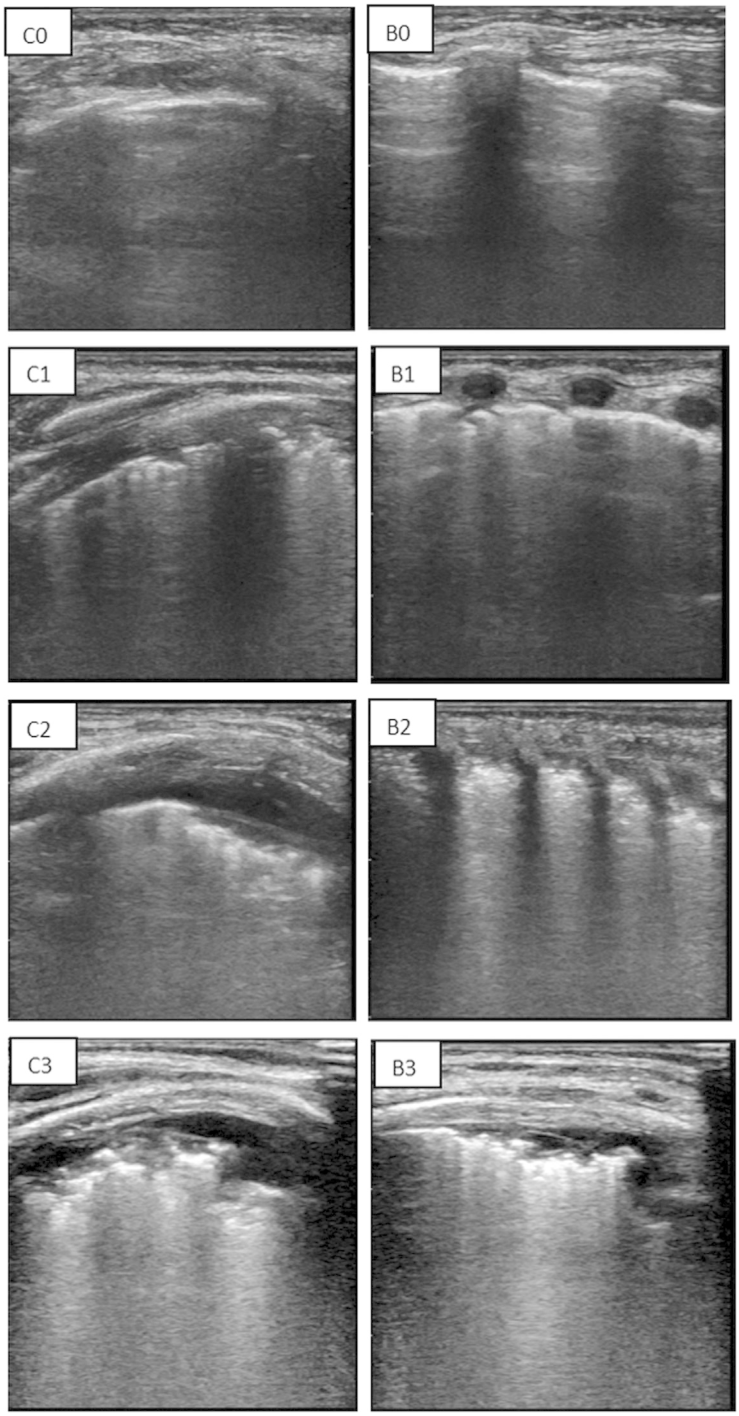

Fig. 2. Scoring system for juxtapleural consolidation and B-lines of lung ultrasound. The degree of juxtapleural consolidation (C): 0 represents no consolidation; 1 represents minimal juxtapleural consolidation; 2 represents small-sized consolidation; 3 represents large-sized consolidation. B-lines (B): 0 represents fewer than three isolated B-lines; 1 represents multiple well-defined B-lines; 2 represents multiple coalescent B-lines; and 3 represents white lung.

both lungs after extubation were lower in Group B (lung ultrasound score: juxtapleural consolidation, $p = 0.01$; B-lines, $p = 0.02$; chest radiography atelectasis score, $p < 0.001$) (Table 4). No statistically significant differences were observed between the two groups in terms of the proportions of patients with lung ultrasound scores of 0, 1, and 2 or chest radiography atelectasis scores of 0, 1, 2, and 4 after extubation ($P > 0.05$ for all comparisons) (Table 4).

The prevalence of hypoxemia (after extubation, $SpO_2 < 90\%$) was 46 (66.7%) in Group A and 14 (20.3%) in Group B ($p < 0.001$) (Table 1). The lowest postoperative SpO_2 within 30 minutes after extubation was better in Group B than in Group A ($p < 0.001$) (Table 2). The SpO_2 level remained normal after nasal cannula oxygen administration at 2 L/min, and the duration of oxygen administration in Group B was shorter than that in Group A after extubation ($p = 0.01$) (Table 1).

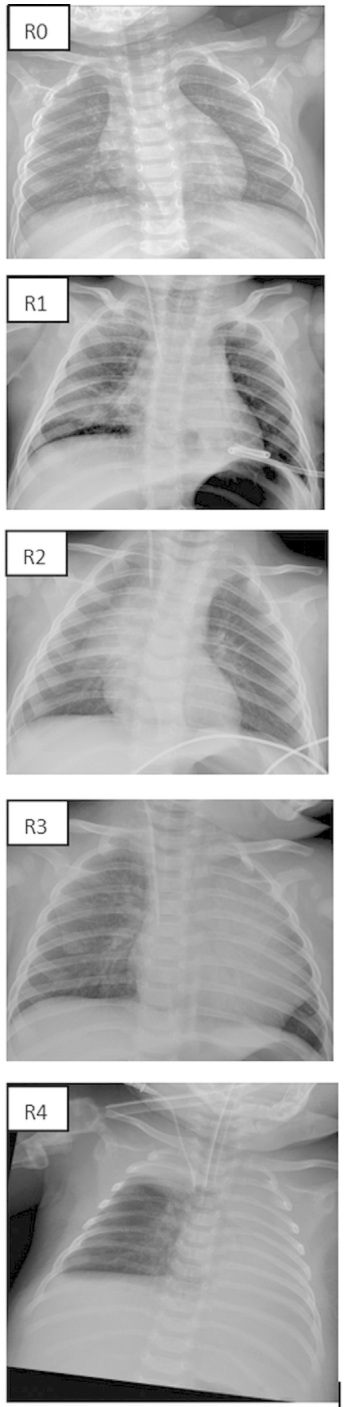

Fig. 3. Scoring system for bedside chest radiography atelectasis (R): 0 represents clear lung fields; 1 represents plate-like atelectasis or slight infiltration; 2 represents partial atelectasis; 3 represents lobar atelectasis; 4 bilateral lobar atelectasis.

Secondary Outcomes

There were no significant differences in the oxygenation index, $AaDO_2$, PaO_2, or SpO_2 at T1 ($p > 0.05$) (Table 2). At T2, the oxygenation index and PaO_2 were greater, and the $AaDO_2$ was lower in Group B than in Group A ($p < 0.001$) (Table 2). The perioperative hemodynamics (MAP, HR, CVP) at T1 and T2 were not significantly different ($p > 0.05$) (Table 3). The time from bronchial blocker removal to extubation was shorter in Group B than in Group A ($p < 0.001$) (Table 1). In terms of the duration of the operation and the length of the postoperative hospital stay between the two groups, there were no significant differences ($p > 0.05$) (Table 1).

Table 1
General Perioperative Conditions of Patients

	Group A (n = 69)	Group B (n = 69)	p
Age (months)	6.0 ± 1.3	5.8 ± 1.4	0.46
Gender (male/female)	34/35	36/33	0.73
Body weight (kg)	6.8 ± 0.9	6.6 ± 1.1	0.29
The duration of operation (min)	98.9 ± 14.2	101.6 ± 13.2	0.27
The prevalence of hypoxemia, n	46 (66.7%)	14 (20.3%)	<0.001
The time from bronchial blocker removal to extubation (min)	12.9 ± 3.2	8.5 ± 2.2	<0.001
The duration of oxygen administration after extubation (min)	61.3±57.8	33.0±67.1	0.01
Postoperative hospital stay (days)	5.4±0.8	5.6±0.7	0.20

Table 2
Intraoperative Respiratory Data

	Group A (n = 69)	Group B (n = 69)	p
PaO_2 (mmHg)			
T1	170.8 ± 11.7	174.3 ± 17.3	0.16
T2	83.1 ± 4.5	96.5 ± 6.1	<0.001
Oxygenation index			
T1	283.3 ± 17.2	284.6 ± 23.1	0.71
T2	271.5 ± 19.5	321.5 ± 20.5	<0.001
$AaDO_2$ (mmHg)			
T1	45.0 ± 4.5	45.5 ± 5.5	0.30
T2	43.7 ± 4.5	37.6 ± 4.5	<0.001
SpO_2 (%)			
T1	98.8 ± 1.0	99.1 ± 1.1	0.13
The lowest postoperative SpO_2 within 30 min after extubation	90.9 ± 3.7	93.2 ± 3.0	<0.001

Table 3
Intraoperative Hemodynamic Data

	Group A (n = 69)	Group B (n = 69)	p
MAP (mmHg)			
T1	64.5 ± 5.3	63.7 ± 5.5	0.39
T2	61.5 ± 4.5	60.5 ± 5.5	0.67
HR (beats/min)			
T1	134.9 ± 14.3	136.6 ± 13.6	0.47
T2	125.5 ± 11.5	126.3 ± 10.3	0.66
CVP (cmH_2O)			
T1	6.3 ± 0.9	6.2 ± 0.8	0.50
T2	7.3 ± 0.9	7.2 ± 1.0	0.78

Table 4,
The Lung Ultrasound Scores and Chest Radiography Atelectasis Scores

	Group A (n = 69)	Group B (n = 69)	p	
Juxtapleural consolidation				
0 score, n	0 (0.0%)	0 (0.0%)	/	0.02
1 score, n	18 (26.1%)	27 (39.1%)	0.10	
2 score, n	45 (65.2%)	42 (60.9%)	0.60	
3 score, n	6 (8.7%)	0 (0.0%)	0.01	
B-lines				
0 score, n	0 (0.0%)	0 (0.0%)	/	0.03
1 score, n	24 (34.8%)	34 (49.3%)	0.08	
2 score, n	40(58.0%)	35 (50.7%)	0.39	
3 score, n	5 (7.2%)	0(0.0%)	0.02	
Chest radiography				
0 score, n	0 (0.0%)	0 (0.0%)	/	0.01
1 score, n	14 (20.3%)	23 (33.3%)	0.08	
2 score, n	39(56.5%)	42 (60.9%)	0.60	
3 score, n	16 (23.2%)	4 (5.8%)	<0.001	
4 score, n	0 (0.0%)	0 (0.0%)	/	

There was a statistically significant correlation between lung ultrasound scores and chest radiographic atelectasis scores in both groups ($p < 0.001$) (Fig 4).

Discussion

The results revealed that, in Group B, the atelectasis scores of both lungs after extubation were lower, and the incidence of hypoxemia was also lower than that in Group A after extubation.

Some scholars have argued that PEEP expands the alveoli and that high concentrations of oxygen enter the alveoli intraoperatively, which might cause absorption atelectasis. Endotracheal intubation removal and PEEP removal, coupled with residual anesthesia drugs and possible wound pain, may aggravate the development of atelectasis.[12-14] In our study, in Group B, not only was the time from bronchial blocker removal to extubation shorter, but the prevalence of hypoxemia after extubation was lower, and at T2, the atelectasis score was better, the oxygenation index and PaO_2 value were higher, and the $AaDO_2$ value was lower. These findings suggest that appropriate PEEP combined with 100% oxygen during emergence from anesthesia after bronchial blockade OLV thoracoscopic lobectomy helps maintain alveolar patency and does not increase the risk of atelectasis while reducing the incidence of hypoxemia during extubation. The risk of perioperative atelectasis in infants is significantly greater than that in adults because of incomplete airway and lung parenchyma development and differences in physiological functions.[15] Patients who receive thoracic surgery have a significantly greater risk of developing pneumonia and atelectasis postoperatively than do those who receive other types of surgery.[16] The application of PEEP helps prevent end-expiratory alveolar collapse, thereby maintaining alveolar surface tension, increasing FRC, and improving lung compliance.[17] Additionally, PEEP enhances the distribution of gases within the alveoli, promoting a

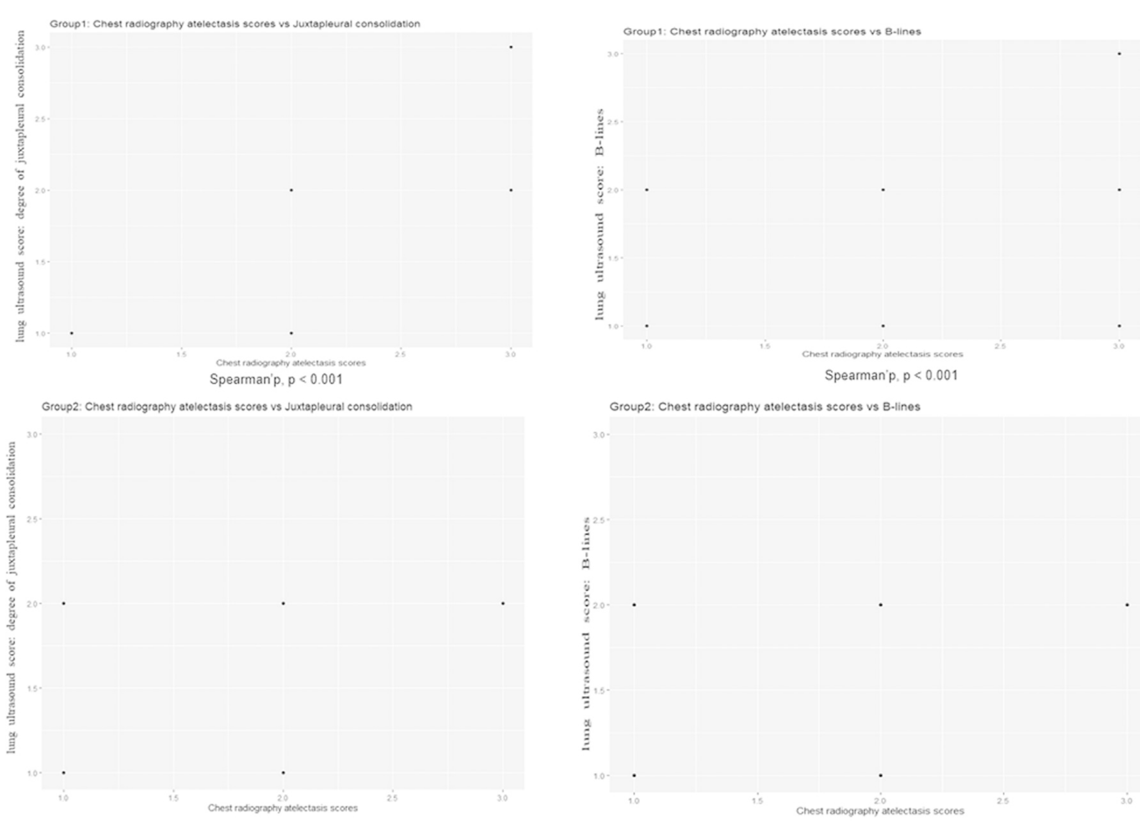

Fig. 4. Correlation between lung ultrasound and chest radiography atelectasis scores after extubation.

more balanced ventilation−perfusion ratio.[18] We also observed that there was no significant difference in SpO_2 between the two groups at T1; however, within the first 30 minutes after extubation, the lowest SpO_2 value was significantly better in Group B than in Group A, and the duration of oxygen administration after extubation in Group B was shorter than that in Group A. This finding indicates that the ventilation strategy used in Group B provided clinically relevant benefits for those patients during the early recovery phase.

Our findings suggested that there was no significant difference in the length of hospital stay between the groups, despite the better oxygenation index and lower atelectasis scores in Group B. These differences might not have been sufficient to significantly prolong the length of hospital stay, and standardized early postoperative care and pulmonary rehabilitation measures might have facilitated a more rapid recovery of respiratory function, further reducing the clinical results of atelectasis.

In contrast, Group A continued with the intraoperative inhaled oxygen concentration but lacked PEEP support during emergence from anesthesia. Consequently, some alveoli might have collapsed at the end of expiration, leading to higher atelectasis scores. Although some patients might have inhaled relatively high concentrations of oxygen, poor ventilation−perfusion matching likely hindered a significant improvement in the oxygenation index.

Erland et al. reported in a randomized controlled trial of nonabdominal surgery in adults who received PEEP of 7 or 9 cmH_2O combined with 100% oxygen during emergence did not increase the incidence of postoperative atelectasis while ensuring sufficient oxygen reserve during extubation, which was consistent with the results of this study.[14] Owing to anatomical and developmental factors, infants have a more compliant chest wall, reduced FRC, and immature alveolar architecture, all of which can influence the response of the lungs to PEEP. Therefore, although the principle may be similar, the application and titration of PEEP in infants must be carefully adjusted according to their unique physiological conditions.

Lung ultrasound has the advantages of being noninvasive and portable and not using radiation; in addition to limiting exposure to radiation, this method is an accurate and reliable method for bedside diagnosis of postoperative atelectasis in children.[19-21] In this study, lung ultrasound combined with bedside chest radiography was used to evaluate postoperative atelectasis. Our results revealed that the two scores were consistent.

If the PEEP level is too low, the alveoli that retract close again, and the effect of retraction cannot be achieved; furthermore, the alveoli open and close repeatedly, resulting in shear injury. This phenomenon has been confirmed in both experimental and clinical studies.[22,23] In contrast, high PEEP levels affect venous return and reduce cardiac output. In addition, high PEEP levels will cause the alveoli to overstretch and easily cause volume injury.[24] The infants in Group B received PEEP of 5 to 6 cmH_2O postoperatively, and the hemodynamics at T1 and T2 were stable without a significant difference, which might indicate that PEEP of 5 to 6 cmH_2O was suitable for thoracoscopic lobectomies with bronchial blocker OLV in infants during emergence from anesthesia.

Limitations

First, the atelectasis scores from both lungs were combined rather than reported separately for the ventilated (operative) and nonventilated (nonoperative) lungs. Although this score reflected the overall lung state after extubation, it might cover lateral differences, especially considering the asymmetric influence of OLV. Second, our research design did not allow us to distinguish the individual effects of PEEP and 100% oxygen because both were applied simultaneously in Group B. Therefore, it was not clear whether the observed benefits were due to PEEP, 100% oxygen, or a combination. Third, this study was a single-center investigation with a small sample size and did not compare different PEEP levels beyond the 5 to 6 cmH_2O range. Future research should include multicenter, large-sample randomized controlled trials to further clarify these findings. In addition, the duration and long-term impacts of the benefits on postoperative oxygenation and lung function should also be evaluated in future studies.

Conclusion

In OLV using bronchial blockers performed during thoracoscopic lobectomies in infants, applying PEEP at 5 to 6 cm H_2O combined with 100% oxygen ventilation after the removal of the bronchial blocker until extubation was associated with improved oxygenation and did not appear to increase the risk of atelectasis during the anesthesia recovery period. However, further studies are needed to evaluate the safety of this strategy.

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Declaration of competing interest

All authors declare that they have no competing interests.

CRediT authorship contribution statement

Ling-Shan Yu: Writing − review & editing, Writing − original draft, Methodology. **Xiu-Hua Chen:** Investigation, Data curation. **Si-Jia Zhou:** Investigation, Data curation. **Zeng-Chun Wang:** Writing − review & editing.

Availability of Data and Materials

The datasets used and analyzed during the current study are available from the corresponding author on reasonable request.

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